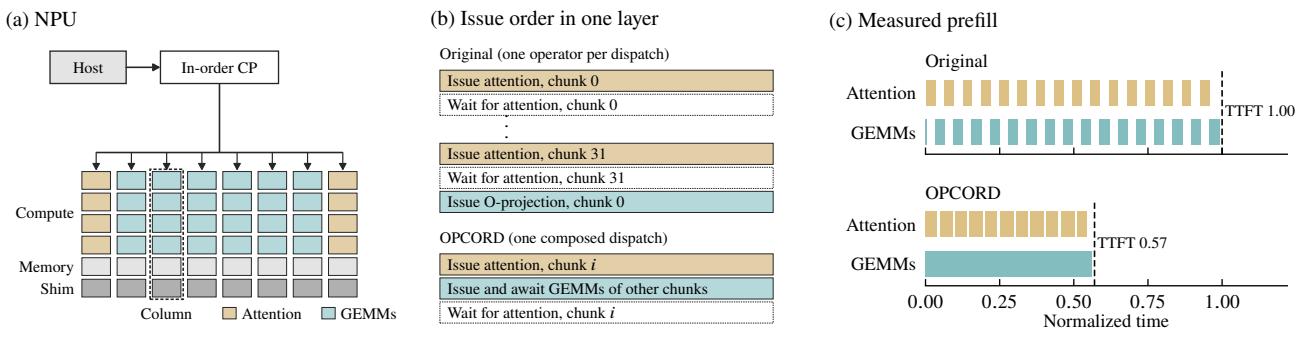

(a) NPU
Host
In-order CP
Compute
Memory
Shim
Column
Attention
GEMMs
(b) Issue order in one layer
Original (one operator per dispatch)
Issue attention, chunk 0
Wait for attention, chunk 0
Issue attention, chunk 31
Wait for attention, chunk 31
Issue O-projection, chunk 0
OPCORD (one composed dispatch)
Issue attention, chunk i
Issue and await GEMMs of other chunks
Wait for attention, chunk i
(c) Measured prefill
Original
Attention
GEMMs
TTFT 1.00
OPCORD
Attention
GEMMs
TTFT 0.57
0.00
0.25
0.50
0.75
1.00
Normalized time